

153. Let the monthly salary of a man be ₹ x .
Then, monthly salary of a woman = ₹ $(x + 500)$.
 $\therefore 4x + 2(x + 500) = 46000 \Leftrightarrow 6x = 45000 \Leftrightarrow x = 7500$.
Monthly salary of a woman = $x + 500 = ₹ 8000$.

154. Let the number of pineapples and watermelons be x and y respectively.

$$\text{Then, } 7x + 5y = 38 \text{ or } 5y = (38 - 7x) \text{ or } y = \frac{38 - 7x}{5}.$$

Clearly, y is a whole number, only when $(38 - 7x)$ is divisible by 5. This happens when $x = 4$.

155. Let F and C denote the temperatures in Fahrenheit and Celsius respectively.

$$\text{Then, } \frac{F - 32}{212 - 32} = \frac{C - 0}{100 - 0} \Leftrightarrow \frac{F - 32}{180} = \frac{C}{100}.$$

$$\text{If } C = 35, \text{ then } F = \left(\frac{35}{100} \times 180 \right) + 32 = 63 + 32 = 95.$$

156. Let the two parts be x and $(74 - x)$.
Then, $5x + 11(74 - x) = 454 \Leftrightarrow 6x = 360 \Leftrightarrow x = 60$.
So, the two parts are 60 and 14.

157. Let the quantity of water drained away be x litres.
Then, $12 - x = x - 6 \Rightarrow 2x = 18 \Rightarrow x = 9$.

158. Let the number of boys in the family be x .
Then each boy has $(x - 1)$ brothers and $(x - 1)$ sisters.
So, number of girls = $(x - 1)$.
Thus each girl has x brothers and $(x - 2)$ sisters.
 $\therefore x = 2(x - 2) \Leftrightarrow x = 2x - 4 \Leftrightarrow x = 4$.
So, there are 4 boys and 3 girls i.e., 4 brothers and 3 sisters in the family.

159. $5000 \text{ INR} = 100 \text{ USD} = \left(100 \times \frac{1}{0.01} \right) \text{ Japanese Yen}$
 $= 10000 \text{ Japanese Yen}$
 $\Rightarrow 1 \text{ INR} = 2 \text{ Japanese Yen}$
 $\Rightarrow 100 \text{ INR} = 200 \text{ Japanese Yen}.$

160. Capacity of the tank = $(13.5 \times 12) \text{ litres} = 162 \text{ litres}$.
 $\therefore$ Required number of buckets = $\left(\frac{162}{9} \right) = 18$.

161. Given exp. = $\left[1 + \frac{1}{1 + \frac{1}{(4/3)}} \right] \div \frac{11}{7} = \left[1 + \frac{1}{1 + \frac{3}{4}} \right] \div \frac{11}{7}$
 $= \left[1 + \frac{1}{(7/4)} \right] \div \frac{11}{7} = \left(1 + \frac{4}{7} \right) \div \frac{11}{7} = \frac{11}{7} \div \frac{11}{7} = 1.$

162. (a) $\frac{1}{3 + \frac{1}{1 + \frac{1}{16}}} = \frac{1}{3 + \frac{1}{(17/16)}} = \frac{1}{3 + \frac{16}{17}} = \frac{1}{(67/17)} = \frac{17}{67}.$

(b) $\frac{1}{2 + \frac{1}{1 + \frac{1}{8}}} = \frac{1}{2 + \frac{1}{(9/8)}} = \frac{1}{2 + \frac{8}{9}} = \frac{1}{(26/9)} = \frac{9}{26}.$

(c) $\frac{1}{3 + \frac{1}{1 + \frac{1}{1 + \frac{1}{8}}}} = \frac{1}{3 + \frac{1}{1 + \frac{1}{(9/8)}}} = \frac{1}{3 + \frac{1}{1 + \frac{8}{9}}}$

$$= \frac{1}{3 + \frac{1}{(17/9)}} = \frac{1}{3 + \frac{9}{17}} = \frac{1}{\left(\frac{60}{17} \right)} = \frac{17}{60}.$$

(d) $\frac{1}{3 + \frac{1}{1 + \frac{1}{2 + \frac{1}{4}}}} = \frac{1}{3 + \frac{1}{1 + \frac{1}{(9/4)}}} = \frac{1}{3 + \frac{1}{1 + \frac{4}{9}}}$
 $= \frac{1}{3 + \frac{1}{(13/9)}} = \frac{1}{3 + \frac{9}{13}} = \frac{1}{(48/13)} = \frac{13}{48}.$

163. Given exp. = $\frac{1}{1 + \frac{1}{3 - \frac{4}{2 + \frac{1}{(5/2)}}}} + \frac{3}{3 - \frac{4}{3 + \frac{1}{(3/2)}}}$
 $= \frac{1}{1 + \frac{1}{3 - \frac{4}{2 + \frac{2}{5}}}} + \frac{3}{3 - \frac{4}{3 + \frac{2}{3}}}$
 $= \frac{1}{1 + \frac{1}{3 - \frac{4}{(12/5)}}} + \frac{3}{3 - \frac{4}{(11/3)}}$
 $= \frac{1}{1 + \frac{1}{3 - \frac{20}{12}}} + \frac{3}{3 - \frac{12}{11}} = \frac{1}{1 + \frac{1}{3 - \frac{5}{3}}} + \frac{3}{(21/11)}$
 $= \frac{1}{1 + \frac{1}{(4/3)}} + \frac{33}{21}$
 $= \frac{1}{1 + \frac{3}{4}} + \frac{11}{7} = \frac{1}{(7/4)} + \frac{11}{7} = \frac{4}{7} + \frac{11}{7} = \frac{15}{7}.$

164. $x = 2 - \frac{1}{1 + \frac{1}{(13/4)}} = 2 - \frac{1}{1 + \frac{4}{13}} = 2 - \frac{1}{(17/13)} = 2 - \frac{13}{17} = \frac{21}{17}.$

165. $x = \frac{2 + \frac{1}{(19/5)}}{2 + \frac{1}{3 + \frac{1}{(5/4)}}} = \frac{2 + \frac{5}{19}}{2 + \frac{1}{3 + \frac{4}{5}}} = \frac{2 + \frac{5}{19}}{2 + \frac{1}{(19/5)}} = \frac{2 + \frac{5}{19}}{2 + \frac{5}{19}} = 1.$

166. Given exp. = $8 - 8 \times \frac{\frac{11}{5} - \frac{9}{7}}{2 - \frac{1}{(35/6)}}$
 $= 8 - 8 \times \frac{\frac{32}{35}}{2 - \frac{6}{35}} = 8 - 8 \times \frac{32}{35} \times \frac{35}{64} = 8 - 4 = 4.$

167. Given exp. = $\frac{2}{2 + \frac{2}{3 + \frac{2}{(11/3)}} \times 0.39} = \frac{2}{2 + \frac{2}{3 + \frac{6}{11}} \times 0.39}$